

Flooding Attack in Energy Harvesting Wireless Sensor Networks

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ABSTRACT

One of the most critical damaging effects on most wireless sensor networks is a flooding attack. Flooding threat occurs not only from spoofed packets generators launched by intruders but also from the inherent behavior of the network protocols. Flooding harm can be categorized into two types as follows. First, a sensor cannot provide required services when it receives a lot of packets. An essential part of packets are rejected and packets delay becomes inadmissible. And second, if a sensor transmits many packets then the sensor battery is quickly exhausted. Energy harvesting technologies offers the potential of reducing the second vulnerability. This paper explores the operation of wireless sensor networks with energy harvesting capability taking into account both flooding negative effects above. We consider the feasibility and threats of the attacks and their impacts. We then propose a theoretical analysis of the flooding attacks. The corresponding mathematical model is offered. Counteracting methods against such intrusions is discussed.

Categories and Subject Descriptors

C.2.0 [Computer-Communication Networks]: General – Security and protection

General Terms

Security, Reliability, Performance.

Keywords

Flooding Attack, Wireless Sensor Network, Energy Harvesting, Energy Consumption

1. INTRODUCTION

The technology of wireless sensor networks (WSNs) has evolved over the last years from a stage where these networks were designed in a technology-dependent manner to a stage where

some broad conceptual understanding issues now exist. An essential progress in wireless communications, micro-electro-mechanical and digital electronics technologies have enabled the development of low-cost, low-power, multifunctional sensor nodes that are attractive for a market. Small wireless sensors, which consist of sensing, data processing, and communicating components, are combined into sensor networks [1]. These networks are intended to be context aware, self-governing, flexible and reliable.

WSNs have a wide range of potential applications, including industrial process monitoring [2], health care [3]-[7], military surveillance [8]-[10], agricultural monitoring [11]-[13], fire detection [14]-[16], smart home [17]. Recent advances in energy harvesting technology for wireless devices have increased the lifetime and performance of wireless sensor node. It allows to create a novel powerful applications. Accordingly the report of GBI Research, the trend of WSNs evolution for the next decade is energy harvesting WSNs. As sensors based applications are deployed, security emerges as an essential requirement. Thus, it is very important to investigate on the potential intrusions against WSNs. It helps us to counteract future threats.

In literature, there are plenty of examples for denial of service attacks in wireless sensor networks [18] – [20]. Cryptography, key management, attack detections and preventions, secure routing, secure location security, secure data fusion, and other security issues in WSNs is considered in depth in [21]. Surveys of attacks against WSNs can be found in [22]. Due to the limited recourses of sensor nodes, one of the most critical damaging effects on most wireless sensor networks is a flooding attack. There are many sources for the material on flooding attack prevention. An efficient protection can be based on path cutoff mechanism, protected paths reservation, packets verification, and packets source authentication [23],[24]. The provided schemes efficiently reject spoofed packets. However, this technique does not protect against the following intrusions. First, a malware at infective node may amplify the transmission range. High illegal electromagnetic emissions cause an exhaustion of a sensor battery. It can also be a reason for noise and degradation of the electromagnetic compatibility of the neighbor sensors. Next, some data aggregation protocols allow to change a packets size. Hence, an intruder can drastically increase a size of legal packets.

The mentioned intrusions lead to quick battery exhausting. Therefore, a sensor node has to increase a time of sleeping mode of energy harvesting mode. It can be a reason of wireless sensor

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node throughput degradation. An offered load for an attacked sensor node has to be reduced. Otherwise, the sensor buffer will be overflowed and packets loss rate will essentially increase. In this situation the effect of flooding attack can be caused by legal packets flow. We consider this situation below.

Remark, while a lot of research has been focused on security issues of wireless sensor nodes equipped by non-rechargeable batteries, the problem of efficient functioning for energy harvesting wireless sensor networks (EH-WSNs) has not been explored enough. Most of the works ignored the energy harvesting facilities as well as the battery charge and discharge processes in WSNs. In this paper we offer an approach, which can fill the mentioned gap.

The rest of the paper is organized as follows. In Section 2 we discuss feasibility of flooding attacks and specific characters of EH-WSNs. Then, we propose a model for the attack effect analysis in Section 3. Some particular protection strategies against the threat are discussed in Section 4. Finally, we conclude the paper in Section 5.

2. CONCEPTUAL FRAMEWORK

It is well known fact that a cost of sensor components is a critical consideration in the design of practical sensor networks. A cost of sensor network increases with sensor battery power. In conventional WSNs it is often economically advantageous to discard a sensor rather than sensor recharging. By this reason a battery power is usually a scare component in wireless devices. On the other hand, sensor lifetime depends on battery lifetime. To increase a sensor lifetime, the concept of duty cycle is implemented. A sensor spends an essential part of lifetime in the sleeping mode (sometimes up to 95%). The main part of sensor energy consumption is caused by packets transmission.

As it was mentioned above, WSNs are vulnerable for flooding attacks. Flooding threat occurs not only from spoofed packets generators launched by intruders but also from the inherent behavior of the network protocols. Flooding harm can be categorized into two types as follows. First, a sensor cannot provide required services when it receives a lot of packets. An essential part of packets are rejected and packets delay becomes inadmissible. And second, if a sensor transmits many packets then the sensor battery is quickly exhausted. Energy harvesting technologies offers the potential of removing this vulnerability.

In EH-WSNs each sensor is supplied with an energy harvesting device. The offered energy sources include solar radiation, wind, vibration etc. Let us remark that the proper operation of any energy-harvesting method is heavily dependent on the environment. For example, the energy harvesting intensity depends on cloud size and wind speed. The recent significant result in this area has been published in [25]. The authors have demonstrated the ability to transfer electric power from one storage device to another without any plugs or wires. Taking into account the feasibility of wireless power transfer it has been offered to apply a mobile wireless charging vehicle for sensor nodes recharging [26]. In this case the energy harvesting intensity depends on the vehicle reliability and recharging schedule. Thus, sometimes in EH-WSNs a sensor node does not harvest energy during long time and it is reasonable to use the sleeping mode.

We consider the following scenario of intrusions. If a sensor transmits packets, an attacker provokes additional energy

consumption for the sensor. The sensor battery is quickly depleted and the sensor node is switched off until a session of energy harvesting. The attack effect can be caused by the following approaches.

- An intruder spreads self-replicating malicious codes. Malware induce a sensor to increase the transmission power.
- An intruder generates relaxed jamming. If a sensor can counteract against a noise by an increase in the power of transmission then, the object of attack is the battery of a sensor [27]
- Some traffic aggregation schemes support the feature of packets fusion. An intermediate malicious node can essentially increase the size of retransmitted packet.

Anyway, a sensor node battery will be recharged. However, the sensor node throughput is depredated. It leads to the flooding attack effect. The packets loss rate is increased.

3. SYSTEM MODEL

3.1 Normal Traffic

First, we first consider the sensor node behavior in a normal situation. Assume, an attack is absent. A sensor node transmits packets, treats information, and sometimes uses a source of energy to recharge the battery. The energy harvesting procedure is a random process. There is no guarantee that the capacity of sensor battery is enough large and a sensor can work from a recharging session to the next session. Assume a firmware of sensors supports the sleeping mode as well. We offer a model of the sensor's behavior based on Continuous Time Discrete States Markov process. Markovian models for energy harvesting process have been proposed in the literature [28], [29]. From our consideration we get the following states of a wireless sensor node.

- State A: A sensor node receives data packets, participates in monitoring, generates commands, treats information, analyzes conditions etc. So, a sensor is active and spends a relatively small account of energy. If a sensor discloses a critical level of energy then the sleeping mode is used.
- State T: A sensor transmits own data or retransmits aggregated data. In this state the energy consumption is high. If the transmission session is completed then a sensor comes back to the previous state and receives a notification of successful transmission.
- State E: A sensor battery is recharged. In this state a sensor is active, but one does not transmits any data.
- State S: A sensor node turns off its sensing and networking circuitry. So, a sensor node spends minute amount of energy.

The state diagram is given in Figure 1. Let the intensity of the sensors deactivating and placing in the sleep mode be γ_A and the intensity of sensors activating is γ_S . Assume a sensor transmits packets with intensity λ_A . An intensity of transmission completion process is λ_T . Let the intensities of energy source activating and recharging be μ and μ_E correspondingly. The index

of state probability corresponds to the state designation. Let us make the following designation, p_S, p_E, p_T, p_A are the steady-state probabilities of sleeping state, energy harvesting state, transmitting state, and processing state correspondingly.

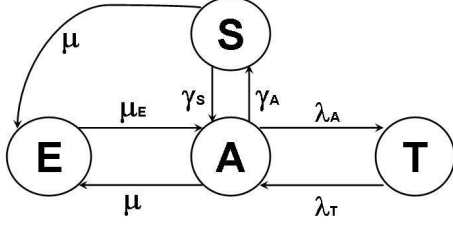


Figure 1. The state diagram for normal situation

From the provided state diagram, we can derive a set of Kolmogorov equations (the balance equations) that can be solved for steady-state occupancy probabilities. We get

$$\begin{aligned} p_S(\mu + \gamma_S) &= p_A \gamma_A, \\ p_T \lambda_T &= p_A \lambda_A, \\ p_E \mu_E &= \mu(p_A + p_S), \\ p_A(\lambda_A + \mu) &= p_E \mu_E + p_T \lambda_T + p_S \gamma_S. \end{aligned}$$

It is a system of four equations in the four variables p_S, p_E, p_T, p_A . Here three equations are linearly independent. Let us take the first three equations. Therefore,

$$\begin{aligned} p_S &= \frac{\gamma_A}{\mu + \gamma_S} p_A, \\ p_T &= \frac{\lambda_A}{\lambda_T} p_A, \\ p_E &= \frac{\mu}{\mu_E} p_A \left(1 + \frac{\gamma_A}{\mu + \gamma_S} \right). \end{aligned}$$

Using the normalization condition,

$$p_S + p_A + p_T + p_E = 1,$$

the probability of state A can be found as,

$$p_A = \left(\Phi_{\mu, \gamma}^{A, S} + \frac{\lambda_A}{\lambda_T} \right)^{-1},$$

where

$$\Phi_{\mu, \gamma}^{A, S} = 1 + \frac{\gamma_A}{\mu + \gamma_S} + \frac{\mu}{\mu_E} + \frac{\mu \gamma_A}{\mu_E(\mu + \gamma_S)}.$$

Hence,

$$p_T = \frac{\lambda_A}{\Psi_{\mu, \gamma, \lambda}^{A, S}},$$

where

$$\Psi_{\mu, \gamma, \lambda}^{A, S} = \lambda_T \Phi_{\mu, \gamma}^{A, S} + \lambda_A.$$

The probability p_T reflects a portion of time that a sensor node spends for packets transmission. In the normal situation, stages of sensor node are balanced. So, sensor nodes are not switched off. The sensor node throughput is enough for required QoS supporting.

3.2 Attack Model

We now consider sensor node stages under the flooding attack. Let α be the attack rate. It reflects the intensity of malefactor traffic. An adversary can create congestions in the network and exhaust sensor resources. If a sensor battery is depleted then a sensor does not work (State OFF) and waits a session of energy harvesting. The state diagram of attacked sensor behavior is given in Figure 2.

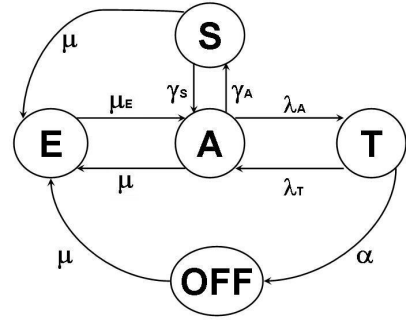


Figure 2. The state diagram for the attack

Let $\tilde{p}_i$ be the probability of state under the attack, where i is the index of a state. Let t be the current time. Let us derive the states probabilities.

The process gets a state i at time $t + \tau$, where τ is a small interval, if one of the following events occurs. First, the system was in state i at the time t , and state is not changed in the interval $(t, t + \tau]$. Next, the system was in an other state j at time t , and the state was changed in the interval $(t, t + \tau]$. The probability of two or more state transitions in the interval $(t, t + \tau]$ is relatively small. The considered events are mutually exclusive and collectively exhaustive. Using the theorem of total probability and taking the limits as $\tau \rightarrow 0$, we get a system of equations for the states probabilities.

The evolution of a the considered process is given by the first-order differential equation

$$\begin{aligned}
\frac{d\tilde{p}_S}{dt} &= -\tilde{p}_S(t)(\gamma_S + \mu) + \tilde{p}_A(t)\gamma_A, \\
\frac{d\tilde{p}_E(t)}{dt} &= -\tilde{p}_E(t)\mu_E + (\tilde{p}_A(t) + \tilde{p}_S(t) + \tilde{p}_{OFF}(t))\mu, \\
\frac{d\tilde{p}_T(t)}{dt} &= -\tilde{p}_T(t)(\lambda_T + \alpha) + \tilde{p}_A(t)\lambda_A, \\
\frac{d\tilde{p}_{OFF}(t)}{dt} &= -\tilde{p}_{OFF}\mu + \tilde{p}_T(t)\alpha.
\end{aligned}$$

In the case of EH-WSNs the state OFF is not absorbing state. Every state is reachable from every other state for the considered continuous-parameter Markov chain. Thus, the considered Markov chain is irreducible. Hence, the stationary state probabilities exist and are independent of an initial state. Therefore, we get steady state balance equations as follows

$$\begin{aligned}
\tilde{p}_S(\gamma_S + \mu) &= \tilde{p}_A\gamma_A, \\
\tilde{p}_E\mu_E &= (\tilde{p}_A + \tilde{p}_S + \tilde{p}_{OFF})\mu, \\
\tilde{p}_T(\lambda_T + \alpha) &= \tilde{p}_A\lambda_A, \\
\tilde{p}_{OFF}\mu &= \tilde{p}_T\alpha.
\end{aligned}$$

Using the normalization condition,

$$\tilde{p}_S + \tilde{p}_A + \tilde{p}_E + \tilde{p}_T + \tilde{p}_{OFF} = 1,$$

we receive

$$\tilde{p}_A = \left(\Phi_{\mu,\gamma}^{A,S} + \frac{\lambda_A}{\lambda_T + \alpha} + \frac{\lambda_A\alpha}{(\alpha + \lambda_T)\left(\frac{1}{\mu} + \frac{1}{\mu_E}\right)} \right)^{-1},$$

Therefore,

$$\tilde{p}_T = \frac{\lambda_A}{\Psi_{\mu,\gamma,\lambda}^{A,S} + \alpha \Phi_{\mu,\gamma}^{A,S} + \lambda_A\alpha \left(\frac{1}{\mu} + \frac{1}{\mu_E} \right)}.$$

Thus, we receive the estimation for sensor nodes throughput under the intrusion.

4. COUNTERACTING

We first demonstrate the effect of the attacks. Let us consider the ratio

$$R = \tilde{p}_T / p_T.$$

It provides an estimation for sensor node throughput degradation. A transmission time of sensors under the attack is R times less than transmission times for the normal traffic. Assume, all intensities above equals 1, except α . Therefore,

$$\Phi_{\mu,\gamma}^{A,S} = 3, \Psi_{\mu,\gamma,\lambda}^{A,S} = 4,$$

and hence $p_T = 0.25$. It means a sensor node has to spend 25% of a time unit for treatment of an offered load. In Figure 3 we show the effect of throughput degradation under the intrusions. From the figure we see that if $\alpha = \lambda_A$ then a sensor node loses more

than 30% capacity (a common used margin of safety). Generally, $\alpha \gg \lambda_A$. Hence, the flooding attack happens.

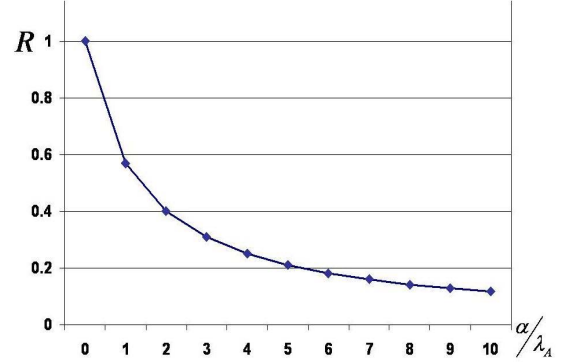


Figure 3. The attack effect

From the formula above we can see that the attack effect is marginal if λ_A is small. It helps us to define an efficient strategy to counteract against the intrusion. The core idea is to decrease an offered load for attacked sensor nodes. Therefore, it is required to detect an intrusion stage. For the detection of the considered intrusion a point-of-change detection technique [30] has to be used. In the following phase, occurring after the attack detection, the attacked sensors environment is notified about the intrusion. The next step includes actions of offered load reducing. It can be as follows

- Paths reassignment. The EH-WSNs packets have to be mainly routed via non-attacked sensor nodes. The sleeping mode of the attacked sensor nodes has to be increased as well.
- Aggressive traffic aggregation. The most critical information is only transmitted. The monitoring findings have to be aggressively merged. The packet size has to be fixed, for example, the Bloom filter can be applied.

The provided defense mechanism can lead to QoS reducing or information quality degradation. But at the same time, the attack effect is essentially reduced.

5. CONCLUSION

In this paper we investigated a survivability of EH-WSNs nodes under the flooding based attacks. Most of the previous works ignored the energy harvesting facilities as well as the battery charge and discharge processes in sensor networks. To fill this gap, the corresponding mathematical models based on stochastic processes have been developed. The offered model allows to estimate defense strategies against flooding attacks in EH-WSNs.

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